

Buckling of Struts

Failure in compression

Short ductile stocky members fail in simple compression due to “Crushing” or “Yielding”. This is due slip, which takes place along planes at roughly 45° to the compressive load. However, long slender beams (usually referred to as Struts) fail under axial compression due to instability.

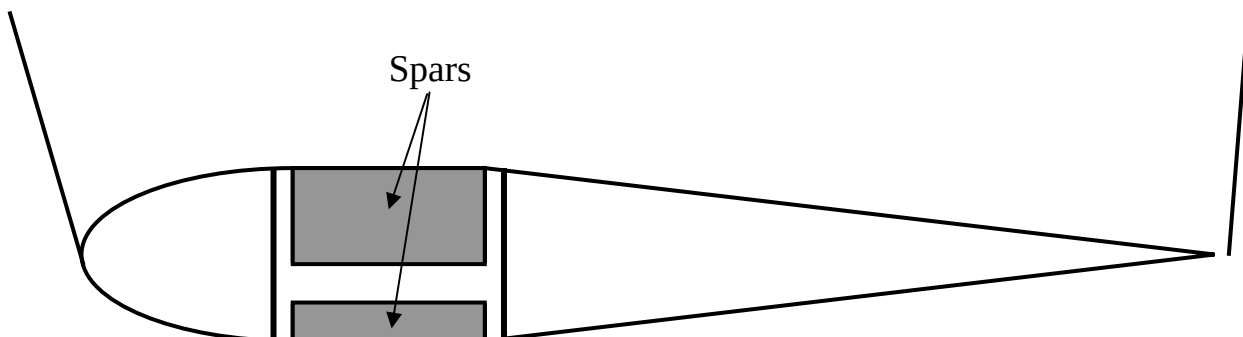
Breaking stresses of materials in Tension and Compression

Tensile strengths are generally widely available for all common engineering materials. However, compressive strengths are generally not widely available because the experimental values of the compressive failure stresses of materials vary a lot with the shape of the test piece. The following table gives some approximate figure for some common materials:

Material	Tensile strength MN/m^2	Compressive strength MN/m^2
WOOD	100	27
CAST IRON	40	340
CAST ALUMINIUM	40	270
ZINC DIE CASTING	35	270
BRITTLE PLASTICS	15	55
CONCRETE	4	40

Based on the above table it may be necessary to design a beam, which has a highly asymmetrical section.

An example of this is the wing-spar of a wooden aircraft, such as a glider, which is always much thicker on the upper or compression side, since wood is weaker in compression than in tension!



What is a Strut

A Strut usually refers to a compression member, which is long in comparison with its cross-sectional dimensions. Such a member will fail due to buckling, well before the direct compressive stress reaches the yield or crushing point. For such an ideal case the strut will remain straight as the load is gradually increased until the critical load is reached when, if then disturbed, it will suddenly buckle. If the load is further increased, the strut will collapse, if it is reduced, the strut will straighten again. The amount of buckling under the critical load is indeterminate. If any of these above Conditions are NOT fulfilled; bending will take place as soon as any load is applied.

Columns or struts subjected to compression are divided into 3 classes:

1. Short struts or columns
2. Intermediate struts
3. Long slender struts

Class 1 Is usually dealt with using yield criteria.

Class 2 Fails due to a combination of crushing and buckling - usually due to slight eccentricity of load. There are modified theories and empirical formulas available that deal with this type of strut.

Class 3 Fails due to elastic buckling. The critical stress can be accurately calculated from using Euler's formula.

Analysis method for different classes of compression members

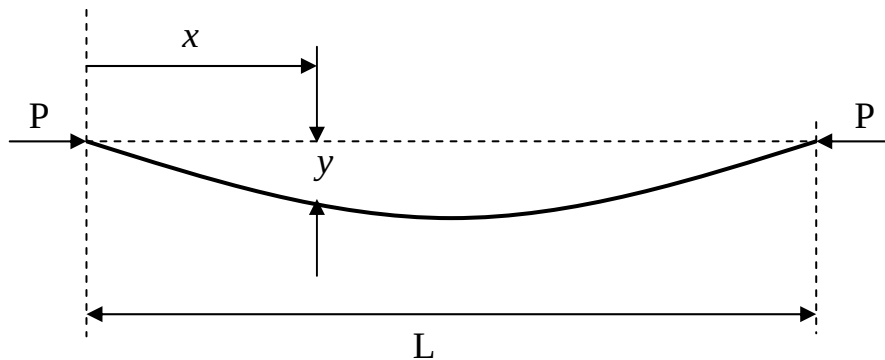
- For a very long strut, where the effect of direct compression is negligible, the load which will cause buckling is given by Euler's theory.
- For shorter struts, the Rankine formula, which makes allowance for the effect of direct compression, is more appropriate.
- For non-linear elastic materials or some aluminium alloys, empirical method such as "Tangent Gradient or Tangent Modulus Method" can be used.
- When dealing with columns which have unsymmetrical cross sections the principal I values must be used.
- For columns where there is stress reversal, alternative method such as reduced E (modulus) method can be employed.
- Other recommended areas of further study by students are; plates buckling, buckling analysis using energy methods etc.

Euler's theory of Buckling for long slender struts with Pin Jointed ends.

Assumptions: -

1. Strut perfectly straight initially
2. Ends free to rotate (Pin Jointed)
3. The load is applied axially.
4. The material is homogeneous.

Let P be the critical or buckling load of the strut. The bending moment M at point 'x' from the LHS, is $M = Py$. From beam deflection theory:



$$-EI \frac{d^2 y}{dx^2} = M$$

$$EI \frac{d^2 y}{dx^2} + Py = 0$$

$$\text{if } \frac{P}{EI} = \omega^2$$

then

$$\frac{d^2 y}{dx^2} + \omega^2 y = 0$$

The solution of this is given by: $y = A \cos \omega x + B \sin \omega x$

At $x = 0, y = 0 \quad A = 0$

Hence $y = B \sin \omega x$

At $x = L, y = 0 \quad 0 = B \sin \omega L$

B cannot be equal to zero otherwise no deflection could occur.

Hence $\sin \omega L = 0$, the solution of which is given by $(\omega L) = 0, \pi, 2\pi, 3\pi$ or

$(\omega L)^2 = 0, \pi^2, 4\pi^2, 9\pi^2$ etc. The smallest value of the P can be found from $\omega L = \pi$ or $(\omega L)^2 = \pi^2$

Or $\omega^2 = \frac{\pi^2}{L^2}$ and for higher modes of failure = $\frac{4\pi^2}{L^2}$ or $\frac{9\pi^2}{L^2}$ etc.

But $\omega^2 = \frac{p}{EI} = \frac{\pi^2}{L^2}$

From which we can find the critical load for the first mode of buckling as;

$$P_{cr} = \frac{\pi^2 EI}{L^2}$$

For convenient length L can be replace with L_e , which is called the effective length of the strut, where $L_e = nL$. Here n is a constant and it is function of the end fixing of the strut.

So for the pin jointed strut the above equation can be written as;

$$P_{cr} = \frac{\pi^2 EI}{(nL)^2}$$

When $n = 1$

Therefore for a pin-jointed strut $L_e = nL$ when $n = 1 \quad \therefore L_e = L$

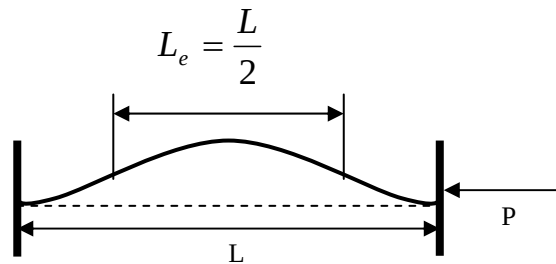
For Pin-Jointed strut failure will occur about the axis which has the least flexural rigidity (i.e. least EI value).

Note that P_{cr} depends on E, NOT upon the ultimate strength of the material!

Effects of end fixing.

- P_{cr} for a strut with both ends fixed is given by :

$$P_{cr} = \frac{\pi^2 EI}{L_e^2}$$



$$P_{cr} = \frac{\pi^2 EI}{(nL)^2}$$

When $n = 1/2$

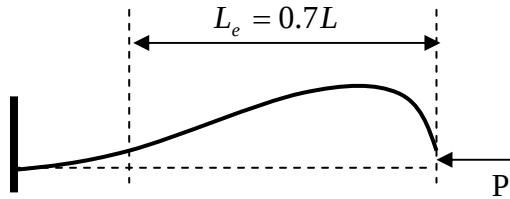
Or

$$P_{cr} = \frac{\pi^2 EI}{\left(\frac{L}{2}\right)^2}$$

Here $L_e = nL$ and $n = \frac{1}{2}$ $\therefore L_e = \frac{L}{2} = 0.5L$

- P_{cr} for strut with one end fixed one end Pined.

$$P_{cr} = \frac{\pi^2 EI}{L_e^2}$$



$$P_{cr} = \frac{\pi^2 EI}{(nL)^2}$$

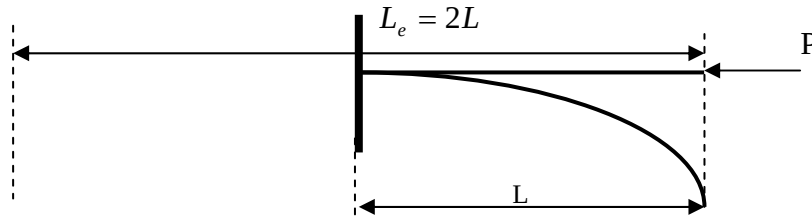
When $n = 0.7$

OR

$$P_{cr} = \frac{\pi^2 EI}{(0.7L)^2}$$

Here $L_e = nL$ and $n = 0.7 \quad \therefore L_e = 0.7L$

- P_{cr} for a strut with one end fixed and the other free :



$$P_{cr} = \frac{\pi^2 EI}{L_e^2}$$

$$P_{cr} = \frac{\pi^2 EI}{(nL)^2}$$

When $n = 2$

OR

$$P_{cr} = \frac{\pi^2 EI}{(2L)^2}$$

Where $L_e = nL$ and $n = 2 \quad \therefore L_e = 2L$

By dividing the critical load by the cross section area and equating that to yield stress one can calculate the critical slenderness ratio that is required for elastic buckling.

Here the slenderness ratio for any given strut, with known property, can be given in terms of critical load or critical stress as;

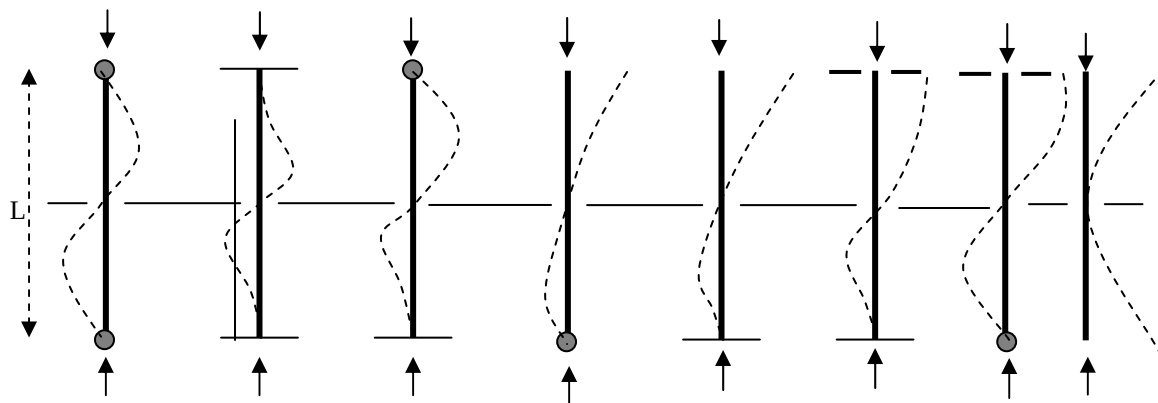
$$\frac{P_{cr}}{a} = \frac{\pi^2 E (ak^2)}{a.L_e^2}$$

$$\sigma_{cr} = \frac{\pi^2 E (k^2)}{L_e^2}$$

$$\frac{L_e}{K} = \frac{nL}{K} = \sqrt{\frac{\pi^2 E a}{P_{cr}}} = \sqrt{\frac{\pi^2 E}{\sigma_{cr}}} \quad \text{where} \quad I = a k^2 \quad \text{and} \quad P_{cr} = \sigma_{cr} \times a$$

Example

Knowing the above four cases, derive an expression for the minimum critical load for each of the following struts that can cause instability or buckling.



$n =$ 1 0.7 0.87 2.7 2.51 1.27 1.4 2

Constraints/side stay — —
Point Loads ↓
Guide — —
Pin Joints ●

Assume solid circular cross section with $d=100\text{m}$, $L=10\text{m}$, $E=200\text{E}9$. Calculate the buckling load in each case.

Rankine Gordon Theory

For a very long strut (class 3), the critical load is given by :

$$P = P_{cr} \text{ (Euler load)}$$

For a very short strut (class 1), the critical load is given by :

$$P = P_y \text{ (load from direct compression and yield).}$$

For a strut of intermediate length (class 2), Rankine's criteria of failure is used which takes into account the Yield strength $P_y = \sigma_y a$ and $I = ak^2$.

$$\frac{1}{p} = \frac{1}{P_y} + \frac{1}{P_{cr}}$$

$$\frac{1}{p} = \frac{P_y + P_{cr}}{P_y P_{cr}}$$

$$P = \frac{P_y P_{cr}}{P_y + P_{cr}} = \frac{P_y}{1 + \frac{P_y}{P_{cr}}}$$

Since $I = aK^2$, where a is the cross section area and K the least radius of gyration.

$$P = \frac{\sigma_y a}{1 + \frac{\sigma_y}{\pi^2 E} \left(\frac{nL}{K} \right)^2}$$

The quantity $\frac{\sigma_y}{\pi^2 E}$ is a constant, called C .

Hence:

$$P = \frac{\sigma_y a}{1 + C \left(\frac{nL}{K} \right)^2}$$

Typical values for C are :

	σ_c (MN/m ²)	C
Aluminium alloy	550	1/1600
Mild Steel	325	1/7500

Intermediate Struts (The Empirical Method)

First we consider the Euler equation. This equation is valid for slender struts with nL/K above validity limit, determined using material properties such as E and σ_{cr} . For intermediate struts Euler's equation must be modified to cater for inelastic buckling. Euler hyperbola is only valid for column having slenderness ratio as small as $\frac{nL}{K}$ since at this point $\sigma_{cr} = \sigma_y$. If the column has slenderness ratio less than $\frac{nL}{K}$, then the critical stress will be greater than σ_y .

If σ_D is a critical stress needed to cause instability we know that in that region strain occurs within a small range so that the modulus of elasticity for the material can be taken as E_t defined as the slope at point D on the stress strain graph. It is as if column behaves as if it is made out of material with lower stiffness than when it behaves elastically, $E_t < E$. As stress increases the **tangent modulus** (the name given to this method) decreases. Using this idea we can modify Euler's equation by substituting E_t for E .

$$P_{cr} = \frac{\pi^2 E_t I}{L_e^2}$$

$I = aK^2$ where "a" is the cross section area. K is the Minimum Radius of Gyration for the section, n the effect due to end conditions. Or $nL = L_e$ the effective length

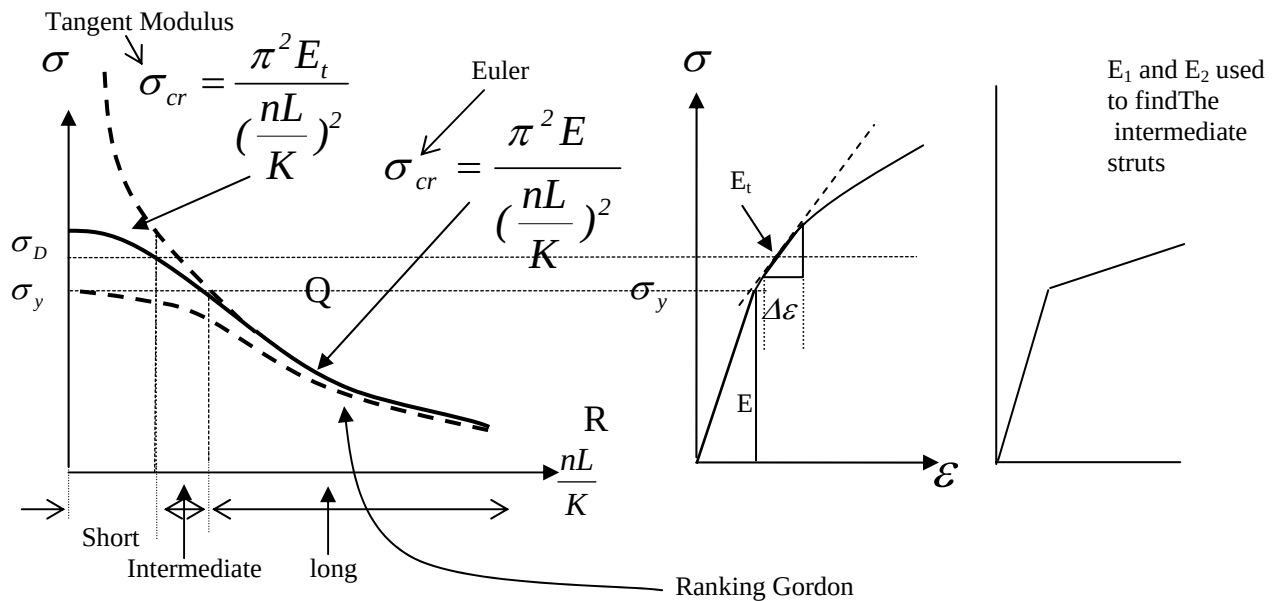
The Euler stress

$$\sigma_{cr} = \frac{P_{cr}}{a} = \frac{\pi^2 E_t (aK^2)}{a(nL)^2} = \frac{\pi^2 E_t}{\left(\frac{nL}{K}\right)^2}$$

$\frac{nL}{K}$ is called 'Slenderness Ratio', and plotting this against σ_{cr} gives the Euler hyperbola as shown below.

Or the slenderness ratio for a give material with known dimension is given by;

$$\text{Slenderness ratio} = \frac{L_e}{K} = \frac{nL}{K} = \sqrt{\frac{\pi^2 E}{\sigma_y}} \quad \text{where } \sigma_y = \sigma_{cr}$$



If a comparison is made with the compressive stress-strain curve of the material. It is seen that for struts of large $\frac{nL}{K}$ in the range Q to R elastic buckling occurs.

Since E is the material property in the Euler formula, it follows that the limit of proportionality must be the limit for which the theory is valid. Various methods have been proposed for bridging the gap between short columns, which fail by yielding, and long slender struts, which fail by buckling. One approach is the Tangent Modulus method, which is particularly suitable for light alloys of types used in aircraft structures, having a gradual stress-strain curve. For intermediate values of $\frac{nL}{K}$, from P to Q, instability will be accompanied by yielding. At a particular point on the stress-strain curve the stiffness is given by E_t , the tangent modulus. It is found that, in this elastic-plastic range, the buckling stress can be predicted using the Euler expression when replacing E by E_t .

In using this method it is useful to have a stress strain curve. By using points on the curve closer together we can almost duplicate the actual stress strain curve. The analysis is done effectively by measuring slope E_t of the curve at various stresses within the elastic plastic range. To get the critical load for a given geometry (L_e/K), we guess a value for E_t and from it determine the stress. Compare this stress with stress associated with that E_t . If a large difference exist, then start again, If not then that load is the critical load/Stress.

Example 1

A straight Spar is 1.2 m long and is simply supported at its ends in a Horizontal position. When loaded at the centre with a concentrated load of 90N, the central deflection was found to be 5mm. If placed vertical and loaded along its axis, what load would cause it to buckle, according to Euler's theory?

Solution

$$y = -\frac{wl^3}{48EI}$$

$$\text{ie } 0.005 = \frac{90 \times 1.2^3}{48EI} \quad \therefore EI = 648 \text{ Nm}^2$$

$$P = \frac{\pi^2 EI}{l^2} = \frac{\pi^2 648}{1.2^2} = 4440 \text{ N}$$

Example 2

A cast aluminium alloy column 200 mm external diameter is 20mm thick and 4.5m long. Assuming that it is rigidly fixed at each end, calculate the safe load by Rankin's formula, Using factor of safety of 4. For Aluminium alloy $\sigma_c = 550 \text{ MN/m}^2$ and $C = 1/1600$.

Solution

$$a = \frac{\pi}{4}(D^2 - d^2) = \frac{\pi}{4}\left(\frac{200^2 - 160^2}{10^6}\right) = 0.01133 \text{ m}^2$$

$$I = \frac{\pi}{64}(D^4 - d^4) = \frac{\pi}{64}\left(\frac{200^4 - 160^4}{10^{12}}\right) = 46.4 \text{ m}^4$$

Since $ak^2 = I$

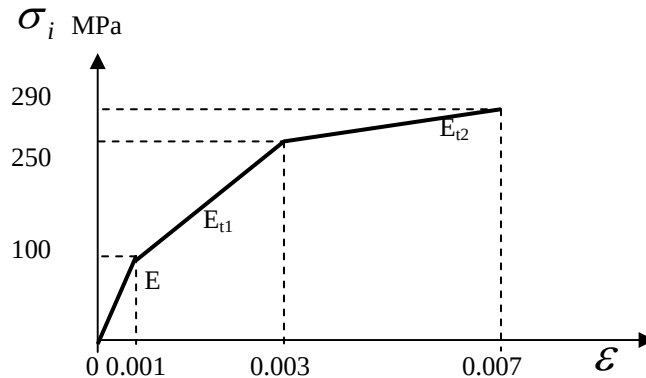
$$\therefore k = \sqrt{\frac{I}{a}} = \sqrt{\frac{46.4 \times 10^{-6}}{0.01133}} = 0.064 \text{ m}$$

From Rankin Gordon theory:

$$P = \frac{1}{4} \times \frac{\sigma_c a}{1 + \frac{C}{4} \left(\frac{l}{k}\right)^2} = \frac{1}{4} \times \frac{350 \times 10^6 \times 0.01133}{1 + \frac{1}{4 \times 1600} \left(\frac{4.5}{0.064}\right)^2} = 879 \times 10^3 \text{ N}$$

Example 3

A vertical under carriage of an aircraft is modeled as a vertical circular column of 76 mm outside diameter and 3 mm thickness. It has a 2m effective length. Determine the axial buckling load for a material with a stress-strain curve approximated as in Figure below.



Solution

For a tube cross section, ID = 70mm & OD = 76mm, we have

$$A = \frac{\pi}{4} (76^2 - 70^2) = 688 \text{ mm}^2$$

$$K = \frac{1}{4} \sqrt{76^2 + 70^2} = 25.8 \text{ mm}$$

Thus $L/r = 77.5$ and the tangent modulus formula is:

$$\sigma_{cr} = \frac{\pi^2 E_T}{(L_e / K)^2} = \frac{\pi^2 E_T}{(77.5)^2} = 0.001643 E_T \quad (a)$$

In as much as the value of E_t is different in every range, we must employ a trial-and-error procedure. For the initial (elastic) range, the slope is $E_t = E = 100 \text{ GPa}$ and Eq. (a) yields ,

$$\sigma_{cr} = 0.001643(100 \times 10^9) = 164.3 \text{ MPa}$$

Since $\sigma_{cr} > \sigma_y = 100 \text{ MPa}$, inelastic buckling occurs. As observed from above Figure, E is not valid above 100 MPa.

From the second portion of the stress-strain curve, $E_t = 75 \text{ GPa}$ and therefore,

$$\sigma_{cr} = 0.001643(75 \times 10^9) = 123.2 \text{ MPa}$$

This is the critical stress at buckling because the value of σ_{cr} falls between 100 and 250 MPa. The solution is therefore

$$P_{cr} = \sigma_{cr} A = (123.2 \times 10^6)(688 \times 10^{-6}) = 84.8 \text{ KN}$$

Tutorials

- 1) An axially loaded aluminium strut of length 6m, with the cross section shown in figure (a) has pin-jointed ends. If $I_{XX} = I_{YY} = 4.70 \times 10^{-6} \text{ m}^4$, $I_{XY} = 2.8 \times 10^{-6} \text{ m}^4$ and assuming that there are no local buckling or rippling effects due to lack of symmetry, determine;
 - The principal second moments of area for the strut section.
 - The relative angle θ between the principal axes and the global XY co-ordinate system.
 - The buckling load if the ends are pin jointed.
 - The buckling load if the ends are built in.
 - If the built in strut is pinned at mid span, restricting its movement in the xy plane as shown in figure b), what is the new buckling load?

Assume Young's Modulus, $E = 72 \text{ GPa}$

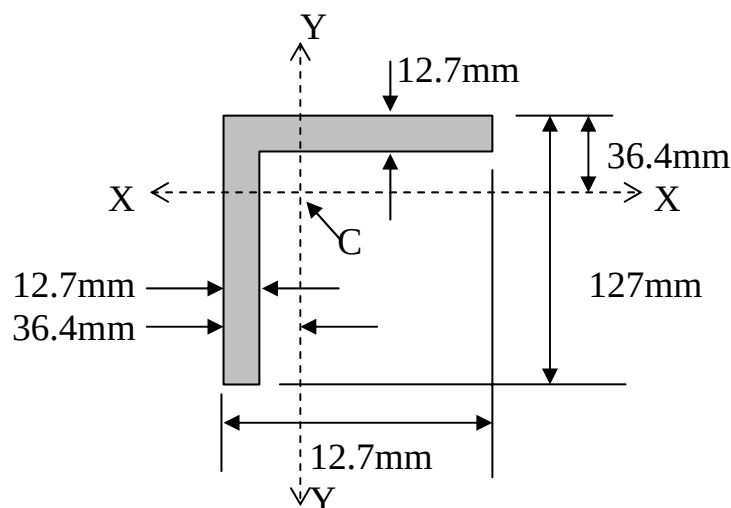


Figure (a)

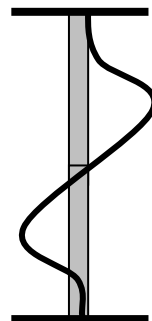


Figure (b)

- 2) An axially loaded aluminium strut of length 6m, with cross section shown in figure (a), has pin-jointed ends. If $I_{xx} = 78E^6 \text{ mm}^4$, $I_{yy} = 26E^6 \text{ mm}^4$ and $I_{xy} = -19.5E^6 \text{ mm}^4$ and assuming that there are no local buckling or rippling effects due to lack of symmetry, determine;
- The principal second moments of area for the strut section I_{uu} , I_{vv}
 - The relative angle θ between the principal axes and the global XY co-ordinate system.
 - The least radius of gyration.
 - The buckling load.
 - If a circular plate, with a round hole in its centre, restricts the middle of the strut in part (a) from moving in the xy plane, as shown in figure (b), what is the new buckling load?

Assume Young's Modulus, $E=72\text{GPa}$

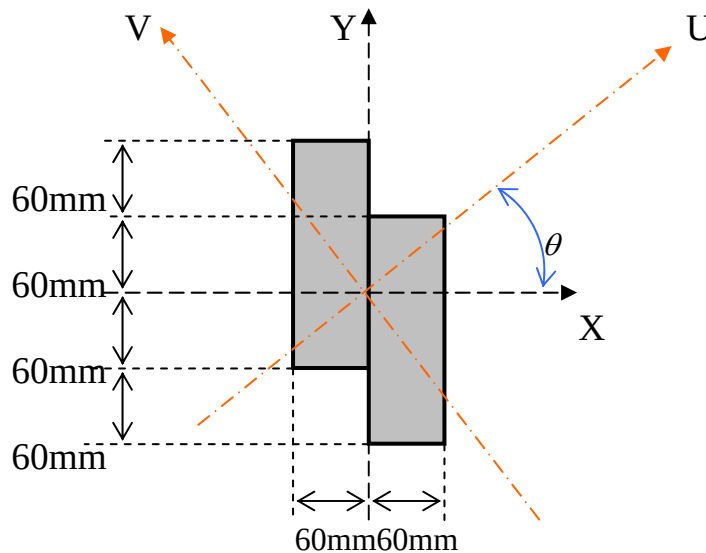


Figure (a)

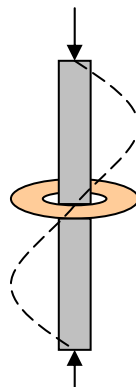


Figure (b)

- 3) An axially loaded composite strut of effective length L , is made by securely bonding together identical planks in each of the arrangements shown in figure (a & b). A single plank is shown in figure (c).

- Calculate the principal second moments of area for the two struts.
- Calculate the buckling load for both arrangements.
- Calculate the ratio of the critical load using arrangement in figure (a) to the critical load using arrangement figure (b).
- If the struts have built in ends and are pinned at mid span, restricting its movement in the xy plane as shown, what is the new buckling load?

Assume Young's Modulus, E 70GPa

PS. Do all your calculations in terms of L and d

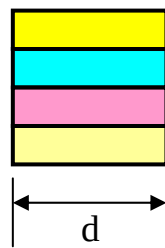


Figure (a)

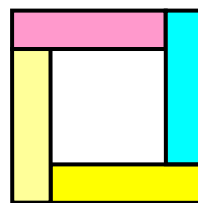


Figure (b)

Figure (c)

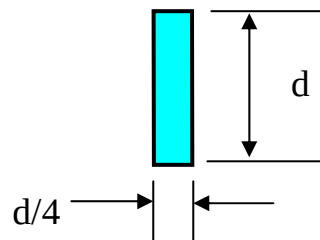
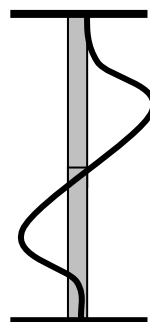


Figure (d)



- 4) An axially loaded aluminium strut of length 6m, with cross section shown in figure (a), has pin-jointed ends. If $I_{xx} = 0.735\text{E}6 \text{ mm}^4$, $I_{yy} = 0.735\text{E}6 \text{ mm}^4$ and $I_{xy} = 0.424\text{E}6 \text{ mm}^4$ and assuming that there are no local buckling or rippling effects due to lack of symmetry, determine;
- The principal second moments of area for the strut section I_{uu} , I_{vv} .
 - The relative angle θ between the principal axes and the global XY coordinate system.
 - The least radius of gyration.
 - The buckling load.
 - If a circular plate, with a round hole in its centre, restricts the middle of the strut in part (a) from moving in the XY plane, as shown in figure (b), what is the new buckling load?

Assume Young's Modulus, $E=70\text{Gpa}$

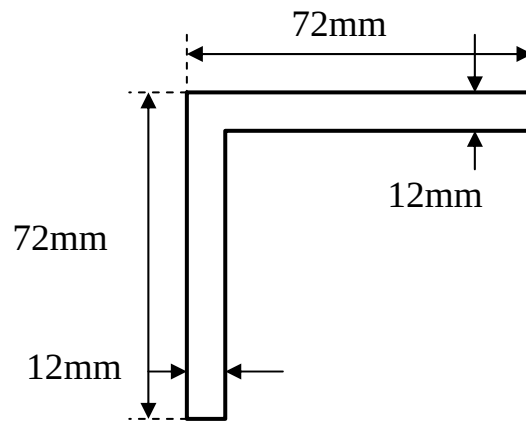


Figure (a)

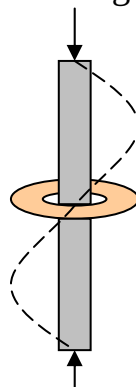
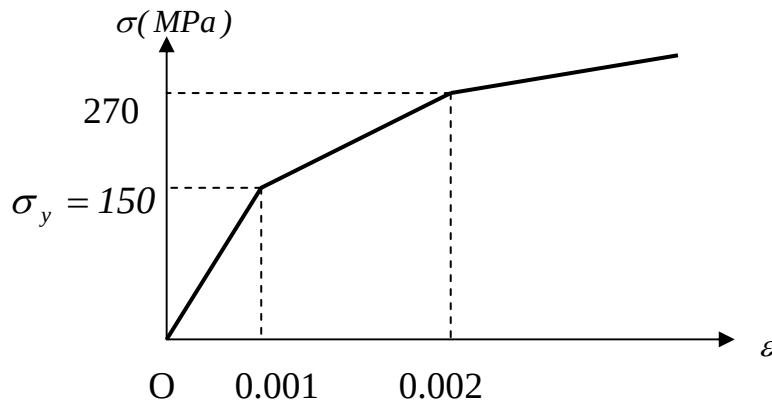


Figure (b)

- 5) A 600mm long solid rod has a diameter of 30mm. It forms part of an aircraft undercarriage and can be modelled as a pin supported column. It is made from a material that can be modelled by the stress-strain curve shown below. Determine the critical buckling load P_{cr} .



- 6) A tubular pillar, external diameter 25cm, internal diameter 22.5cm and length 10m, is subjected to an eccentric load of 20tonne, as shown below. Calculate the maximum permissible eccentricity if the maximum permissible stress is 75MPa. $E = 70$ GPa.

(Ans, A=119mm & B= 83.1mm)

